

Transformations of $f(x) = |x|$, Part II - Practice

Algebra II Honors | Unit 2 | Lesson 3 Practice | TEK A2.6.C

Objective: I can factor a combined expression into vertex form, and work backward from a vertex and a point to find a.**Section 1 — Recall**

1. Factor $f(x) = 4|2x - 8|$ into vertex form $a|x - c|$, and give the vertex.

$$4|2(x - 4)| = \underline{\hspace{2cm}} \quad \text{Vertex: } \underline{\hspace{2cm}}$$

2. A function has vertex (5, -3) and passes through (7, 1). Find a.

$$a|7-5| - 3 = 1 \rightarrow 2a = \underline{\hspace{2cm}} \rightarrow a = \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Factor $f(x) = 3|6x + 18| - 2$ into vertex form, and give the vertex and real a-value.

$$3|6(x + 3)| - 2 = \underline{\hspace{2cm}}$$

$$\text{Vertex: } \underline{\hspace{2cm}} \quad a = \underline{\hspace{2cm}}$$

4. A function has vertex (-1, 4) and passes through (2, -5). Find a and write the equation.

$$a|2-(-1)| + 4 = -5 \rightarrow 3a = \underline{\hspace{2cm}} \rightarrow a = \underline{\hspace{2cm}} \quad f(x) = \underline{\hspace{2cm}}$$

Section 3 — Challenge

5. A function has vertex (3, 2) and passes through BOTH (1, 8) and (5, 8). Verify both points give the same value of a.

From (1,8): $a = \underline{\hspace{2cm}}$ From (5,8): $a = \underline{\hspace{2cm}}$ (both points are the same distance from the vertex)

6. Derive the general formula: if $f(x) = a|x - c| + d$ passes through (p, q) where $p \neq c$, then $a = (q - d)/|p - c|$.

$$a|p - c| + d = q \rightarrow a|p - c| = q - d \rightarrow a = \underline{\hspace{2cm}} \quad (\text{valid since } p \neq c \text{ means } |p - c| \neq 0)$$

Section 4 — Explain It

7. Explain why knowing a function's vertex and just ONE additional point (not two) is always enough to find a, using the structure of vertex form.
